Studying toxic effects of Alzheimer's risk variant in *CASP8* using cellular models

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Abstract

Alzheimer's disease (AD) impacts 7 million + Americans and is the most common form of dementia. AD is a multifactorial disease with its genetics not fully understood. Neuropathological features of AD include hyperphosphorylation of the tau protein (pTau) and accumulation of amyloid-β plaques. Our lab recently discovered a repeat expansion mutation, $GGGAGA^{exp}$, encoding poly-glycine-arginine (polyGR) in *CASP8*. A repeat variant of this expansion, *CASP8*-AD-R1, has been linked with increased AD risk with an ~2.2 odds ratio. Our lab also found polyGR linked to AD neuropathological changes, e.g. increasing pTau levels in an overexpression cell culture model. Here, we aim to investigate the relationship between polyGR produced by *CASP8*-$GGGAGA^{exp}$ and pTau using neuronal cultures. Differentiated SH-SY5Y cells, which have been widely used to model neurons, were divided into three groups: the first group received AAV-AD-R1 virus; the second group received AAV-C-Var virus, a more frequently seen variant containing $GGGAGA^{exp}$, (*CASP8*-C-Var); and a control group receiving AAV-Cntl virus containing $GGGAGA_3$. SH-SY5Y cells were differentiated with retinoic acid (RA), transduced with AAVs, then stained via immunofluorescence (IF) staining. Brightfield imaging and IF results confirmed the efficacy of the differentiation and transduction protocol. Qualitative analysis revealed notable co-localization between polyGR and pTau in cells that received AAV-AD-R1 and AAV-C-Var compared to AAV-Cntl. In summary, these results show that differentiated SH-SY5Y cell models transduced with AAV viruses expressing disease-associated repeat expansion variants could be an effective model to further study AD pathology, polyGR, or other repeat expansion mutations related to AD.

Keywords: Alzheimer's Disease, repeat expansion mutations

Studying toxic effects of Alzheimer's risk variant in *CASP8* using cellular models

Alzheimer's disease (AD) is a neurodegenerative disease affecting ~7 million Americans age 65 or older and is the cause of 60-80% of dementia cases. The worldwide population with dementia is expected to reach 131.5 million by 2050, making AD an increasing worldwide issue (Prince et al., 2015). Currently, there is no approved treatment to reverse or arrest the progression of AD (van der Kant et al., 2020).

AD is a progressive neurodegenerative disorder characterized by neuropathological changes (AD-NCs) including the aggregation of amyloid-β (Aβ) plaques and the hyperphosphorylation of the tau protein (pTau) resulting in neurofibrillary tangles (Breijyeh et al., 2020). Aβ results from the sequential cleavage of the amyloid precursor protein (APP) by β-secretase and γ-secretase, producing Aβ40 and Aβ42. Of the two forms of Aβ produced from this pathway, Aβ42 is considered the most toxic due to its C-terminus having more structured hydrophobicity, giving it greater propensity to form amyloid plaques (Chen et al., 2017). Neurofibrillary tangles are composed of hyperphosphorylated pTau that has altered binding to microtubules and increased aggregation propensity. Interestingly, prior research has shown that pTau could result from an upstream accumulation of Aβ *in vivo*, yet other studies have suggested pTau and Aβ work interdependently to cause AD-NCs *in vitro* cultured neurons (Bloom, 2014). These results highlight incomplete understanding in the molecular mechanisms of AD.

Mutations in familial genes of Alzheimer's disease include *APP*, *PSEN1*, and *PSEN2*, which all result in phenotypical changes of an increased Aβ42/Aβ40 ratio and formation of plaques, with *PSEN1* being able to identify up to 70% of familial onset AD cases (Tanzi, 2012 & Andrade-Guerrero, 2023). However, only ~1% of AD cases are caused by mutations in either of these familial genes, inheritance of genes at other loci is not fully understood, and spontaneous

onset of AD takes up to 95% of all cases (Tanzi, 2012). Therefore, other genetic variants that have not been well explored in AD genetics studies, like repeat expansions, could be able to explain this gap in knowledge.

Repeat expansion mutations are sequences of tandem nucleotide repeats within the genome that as expanded can cause disease or increased disease risk. The pathogenesis of repeat expansion disorders includes multiple pathways, ranging from epigenetic transcription silencing to RNA gain of function to protein loss of function. Additionally, through repeat-associated non-AUG (RAN) translation, repeats can be translated into poly-amino-acid proteins that can form protein aggregates. There are growing number of studies that report toxicity of RAN proteins in disease models of repeat expansion disorders (Malik et al., 2021). Because a $GGGGCC^{exp}$ expansion C9 ALS/FTD can cause the accumulation of RAN proteins (Tao et al., 2015), our lab investigated if RAN proteins are also present in AD autopsy brain tissue.

Two studies using postmortem AD brains found a significantly increased amount of poly-glycine-arginine (polyGR) aggregates in the hippocampus; these aggregates were found to have a positive correlation with Aβ and pTau (Nguyen et al., 2025 and Phuong and Tomas et al., 2025). Using polyGR+ aggregate signature, our lab identified a tandem repeat expansion of polyGR-encoding $GGGAGA^{exp}$ in the caspase 8 gene on chromosome 2 (*CASP8*-$GGGAGA^{exp}$) within an SVA-E element. SVA elements are hominid-specific retrotransposons containing tandem repeats; the E family is human-specific and one of the most recent evolutionarily classes of SVA that are not yet well studied (Wang et al., 2005). The *CASP8*-$GGGAGA^{exp}$ was shown to produce polyGR containing proteins using minigenes and overexpression cells. Additionally, Nguyen et al., 2025 found that presence of these aggregates led to increased pTau levels *in vitro*. Furthermore, association studies using cohorts of AD cases and age-similar controls revealed a

specific sequence variant of *CASP8*-GGGAGA^{exp} (C8-AD-R1) that was significantly linked to AD with an odds ratio of ~2.2 (Nguyen et al., 2025). These results are exciting because they link the repeat expansion variant to AD and raise an important question on how this variant contributes to neuropathological changes and risk. Building on the correlation of pTau and polyGR in AD autopsy brains and in transfected cells, we hypothesize that expression of C8-AD-R1 using an adeno-associated virus (AAV) approach in differentiated human neuroblastoma (SH-SY5Y) cells will lead to increased pTau levels and contribute to cytotoxicity.

Methods

We utilize a cellular model of SH-SY5Y cells to observe for pathological impacts of C8-AD-R1. This involves differentiating the SH-SY5Y to become mature neuron-like cells, and in the process transducing them with an AAV, then performing immunofluorescence staining (IF) and Western blot on the proteins of interest.

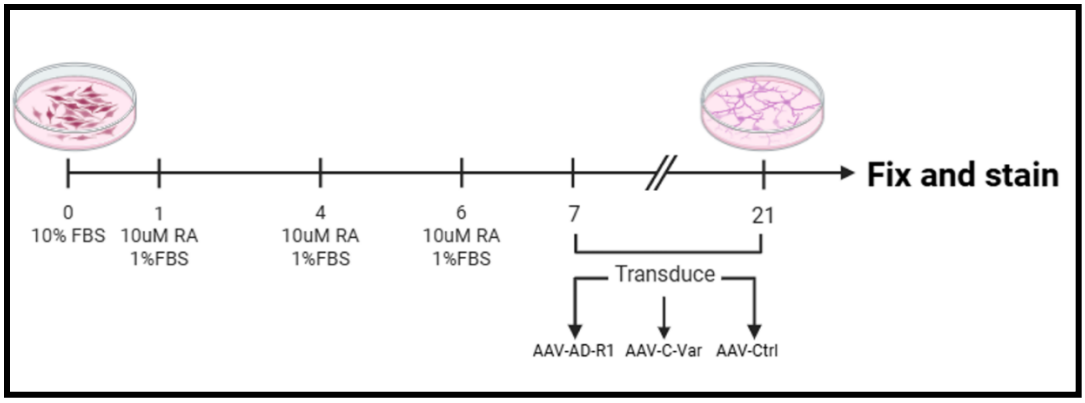
Cell culture

Neuroblastoma cell culture

SH-SY5Y cells are an immortalized cell line. The cell line is placed in an incubator at 37 °C with 5% CO_2 for the duration of the experiments. SH-SY5Y cells were counted using a Bio-Rad® TC20 Automated Cell Counter. Following cell count, 80,000 cells were seeded in 12 well plates for differentiation. SH-SY5Y cells differentiation and transduction is a 21-day process and is divided into three steps: initial differentiation, viral transduction, and expression. Figure 1 illustrates the experimental design, and a description of each step is listed below.

Figure 1

Differentiation protocol of SH-SY5Y cells utilizing 10 μM retinoic acid (RA) and 1% Fetal Bovine Serum (FBS)



Note. Days 1, 4, and 6 mark when differentiation medium is changed. Day 7 marks the transduction of cells with adeno-associated viruses (AAVs). Day 21 marks the end of the expression and start of fixation.

SH-SY5Y differentiation

Following a modified protocol from Targett et al., 2024, cells were differentiated with retinoic acid. SH-SY5Y have been observed to increase synaptic and dendritic growth and receptor specificity, resembling mature neurons. For the first 7 days, at day 1, 4, and 6, cells were given medium with 10 µM of retinoic acid (RA) and 1% Fetal Bovine Serum (FBS), the latter providing necessary nutrients and growth factors to stimulate growth. At day 7, cells were transduced with AAV-containing medium. Every 2 days in the following 14-day period, brightfield images of cells were taken 20x magnification under a Leica® DM IL LED microscope to observe cell growth and progression. Data was collected in a qualitative, observation-based manner. Representative images of each well of cells were chosen to display in results. During this period, medium was changed every 2 to 3 days.

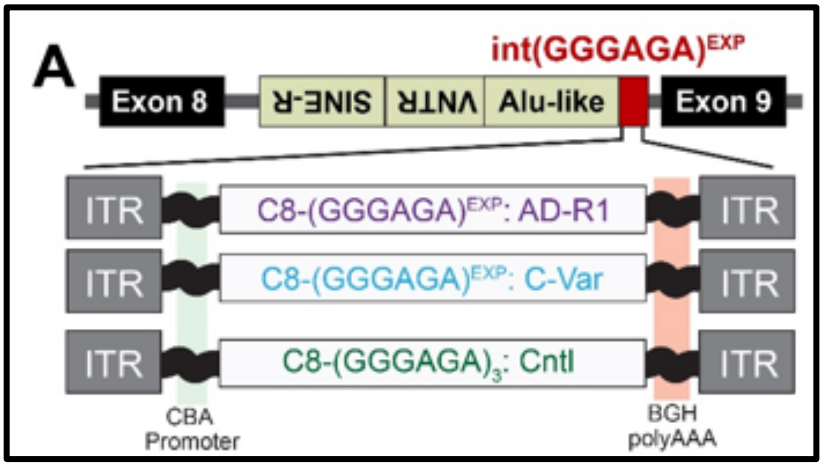
Transduction

Alongside to C8-AD-R1, Nguyen et al., 2025 found a common repeat variant (C-Var) not shown to increase AD risk. Differentiated SH-SY5Y were transduced with either AAV-AD-R1,

AAV-C-Var or the control AAV (AAV-Cntl) (Nguyen et al., 2025). The AAV-Cntl carried a small number of repeats and was not expected to produce polyGR aggregates. As seen in Figure 2, adeno-associated viruses (AAVs) containing $GGGAGA^{exp}$, resembling AD-R1, C-Var, and Cntl were used to transduce the differentiated SH-SY5Y on day 7. The concentration of AAV was determined by both the seeded cell count and the virus' multiplicity of infection (MOI). MOI is the ratio between the number of viral genomes (vg) added to the number of target cells. For our AAV, an MOI of 10,000 vg/cell was used.

Figure 2

Construct diagrams of AAV-AD-R1, AAV-C-Var, and AAV-Cntl



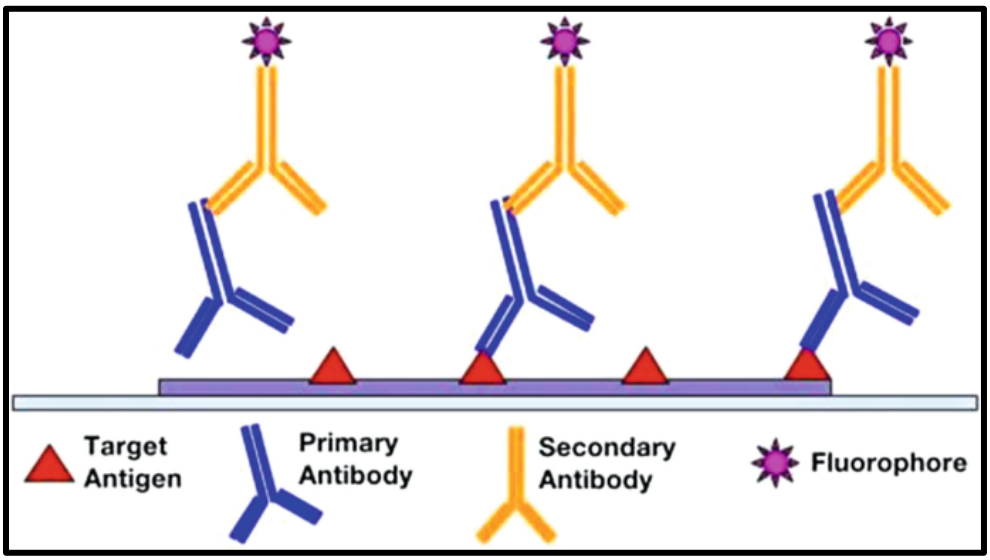
Note. Figure provided by R. Tomas (personal communication, July 17, 2026). Used with permission.

Immunofluorescence staining

Indirect immunofluorescence (IF) staining was performed over a 2-day timespan. Day 1 involved fixation, permeabilization, blocking, and attachment of primary antibodies. Day 2 involved attachment of secondary antibodies. A diagram of IF staining is shown in Figure 3.

Figure 3

Diagram of indirect immunofluorescence (IF) staining



Note. Primary antibodies attach to a target antigen; secondary antibodies containing fluorophores attach to the primary antibodies. Reprinted from *An Introduction to Performing Immunofluorescence Staining*, by Im et al., 2019, *Methods in molecular biology (Clifton, N.J.)*, *1897*, 299–311 (https://doi.org/10.1007/978-1-4939-8935-5_26).

Fixation, permeabilization, and blocking

Cells were fixed using 4% paraformaldehyde (PFA) for 20 min at room temperature (RT). Cells were permeabilized with 0.5% TritonX diluted in phosphate buffer solution (PBS) for 20 min at RT. Cells were blocked with 1% Normal Goat Serum (NGS) diluted in PBS for 1 h at RT. Cells were washed 3x with PBS between each of the steps above.

Primary antibodies

Primary antibodies served three purposes. First, to stain for mature neuronal markers, anti-microtubule-associated protein 2 (MAP2) and anti-βIII-tubulin (Tuj1) antibodies were used. To further validate differentiation, excitatory and inhibitory neurons were stained (VGlut and GABA). Second, anti-polyGR (polyGR) antibodies were used. Third, anti-pTau (AT8) antibodies were used. Antibodies were diluted in 1% NGS in a 1:1000 ratio. 300 µL of solution were added per well. Primary antibodies were incubated overnight at 4 °C. A list of primary antibodies used can be seen in Table 1.

Table 1

List of primary antibodies

Antibody	Targeted protein	Host animal	Function	Color in figures
AT8	Hyperphosphorylated tau protein	Mouse	AD-NC marker	Red
GABA	γ-aminobutyric acid	Mouse	Neurotransmitter marker	N/A[1]
MAP2	Microtubule-associated protein 2	Chicken	Mature neuronal marker	Turquoise
polyGR	Poly-glycine-arginine aggregates	Rat	polyGR marker	Green
Tuj1	βIII-tubulin	Rabbit	Early neuronal marker	Orange
VGlut	Vesicular glutamate transporter	Mouse	Neurotransmitter marker	Violet

Note. Insufficient images were taken for GABA staining to be displayed in results.[1]

Secondary antibodies

Secondary antibodies were prepared in the dark at RT in 1% NGS solution. After 3x wash with PBS, 300 µL of antibody solution were added per well. Cells were incubated for 1 h at RT, washed 3x with PBS, then mounted onto slides containing 1 drop of VECTASHIELD® PLUS

Antifade Mounting Medium with DAPI. After sitting at RT for 1 h, slides were stored at 4 °C. A list of secondary antibodies used can be seen in Table 2.

Table 2

List of secondary antibodies

Targeted animal	Fluorophore wavelength (nm)	Color (naked eye)
Chicken	647	Dark red
Rabbit	555	Orange
Rat	488	Green
Mouse	594	Red

Confocal microscopy

Slides were imaged using a ZEISS® LSM 780 confocal laser scanning microscope mounted on a ZEISS® Axio Observer.Z1 inverted microscope stand in 40x and 63x magnification through oil. Colors were assigned to each primary antibody, as seen in Table 1. 10 images were taken for each slide. Representative cells from those images were chosen to be displayed in results. Note that due to time constraints, cells co-stained with GABA and Tuj1 and C-Var and control cells co-stained with polyGR and Tuj1 were not imaged.

Results

Differentiation

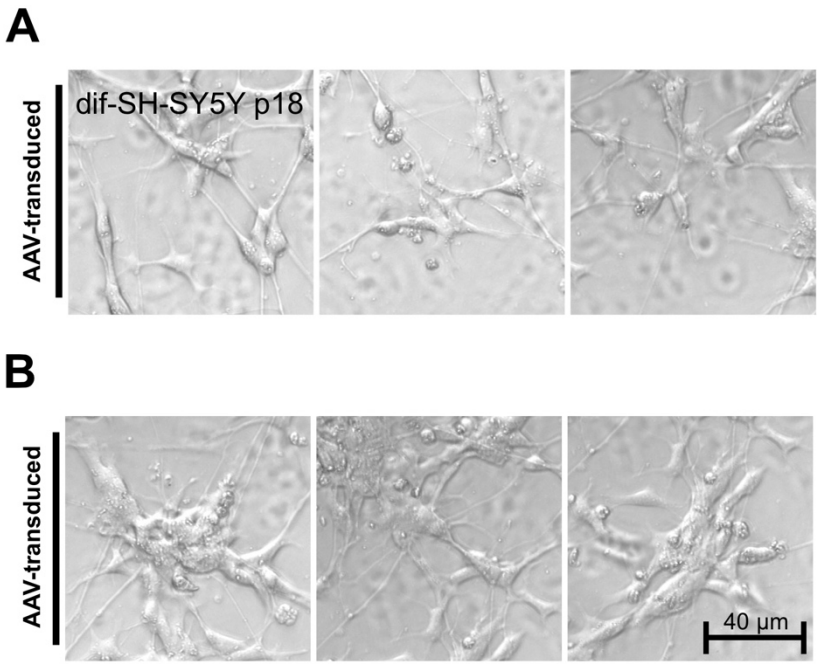
Morphology of cells during differentiation

Brightfield images show marks of initial differentiation. In general, cells were observed to have grown in axonal length, formed clusters, and have increased branching. Representative

images of cells during the differentiation process can be seen in Figure 4. Overall, signatures of maturation are observed; however, we must use mature neurons markers to confirm if they have reached the mature state.

Figure 4

Brightfield images of transduced SH-SY5Y cells



Note. Images taken under 20x magnification. Cells were differentiated for 18 days. Cells were AAV-transduced. This specific representative set was AAV-C-Var. Images represent branching (A) and clustering (B) morphology.

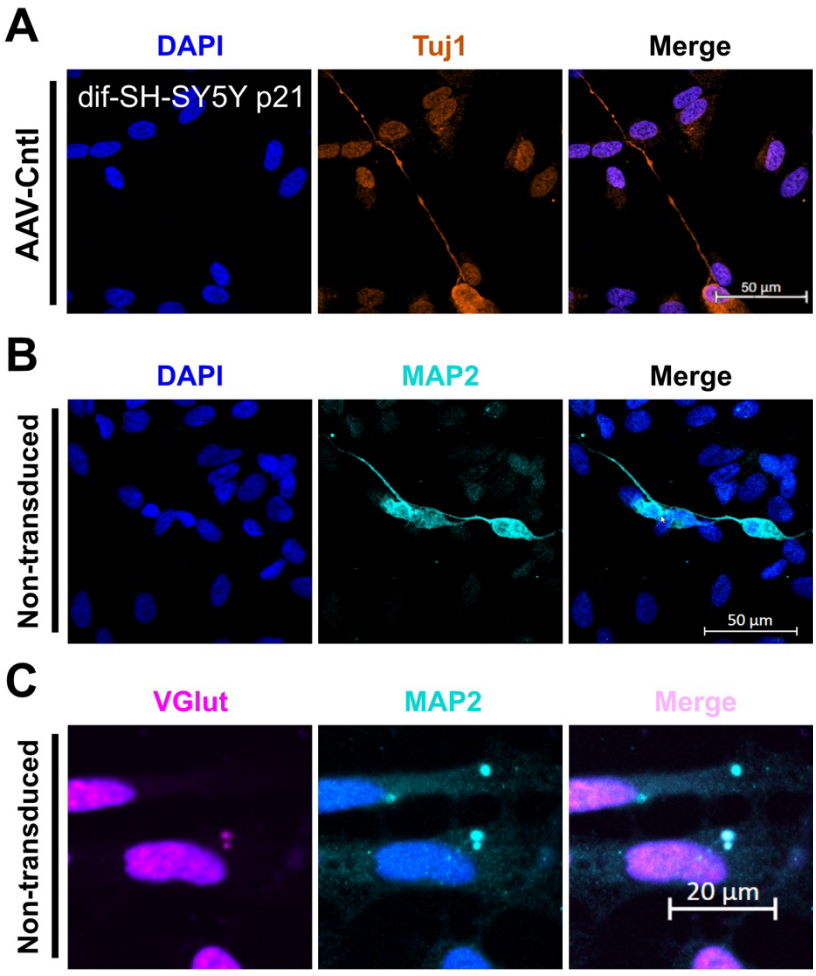
Immunofluorescence staining for mature neuronal markers

On 6 well plates of non-AAV-transduced differentiated SH-SY5Y cells, MAP2 and VGlut was IF stained and imaged. Qualitative observations revealed significant MAP2 expression in our differentiated SH-SY5Y cells, along with some glutamate expression. Glutamate was

observed expressed in the cell body, close to the nucleus, and on the edges of the cell. This can be seen in Figure 5.

Figure 5

Immunofluorescence staining of mature neuronal markers



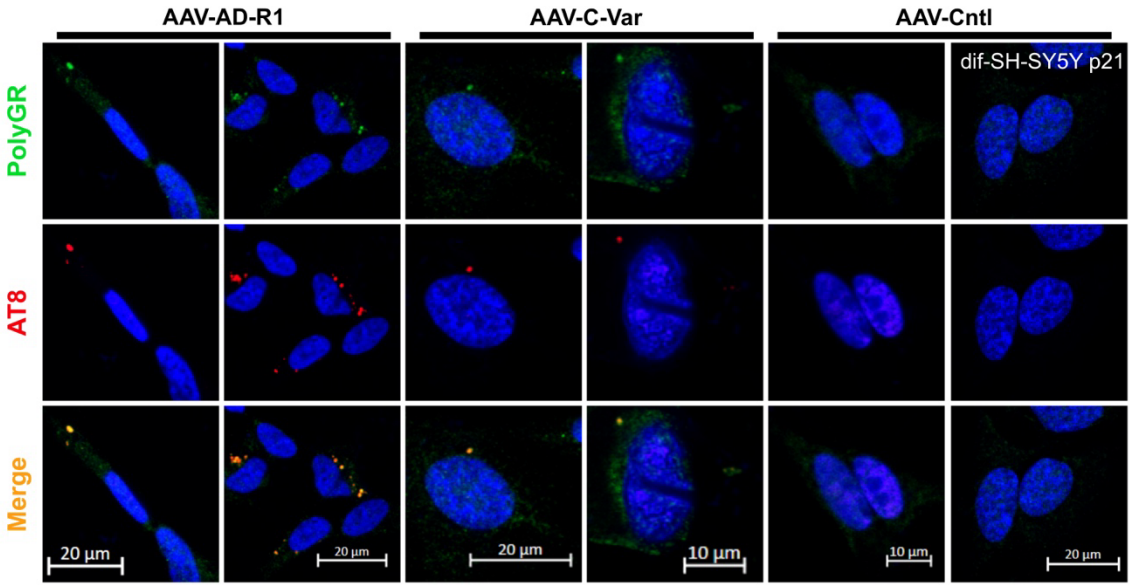
Note. Images taken under 40x magnification. Cells were differentiated for 21 days. Representative images of AAV-Cntl transduced Tuj1-stained control group and non-transduced MAP2-stained cells were chosen (A); Representative images of non-transduced VGlut and MAP2 co-stained cells were chosen (B-C).

PolyGR and AT8 staining

Next, we wanted to investigate the relationship between polyGR, the product of GGGAGA^{exp}, and AD-NCs. Therefore, we stained using polyGR and AT8 antibodies. C8-AD-R1-expressing cells produced polyGR aggregates and pTau, with some but not all co-localizing. C8-C-Var-expressing cells too produced polyGR aggregates and pTau, with some but not all co-localizing, and visually less than the AD-R1 group. Control cells had very little to no expression of polyGR and pTau. Almost none observed showed co-localization of polyGR and pTau. This is shown in Figure 6.

Figure 6

Immunofluorescence co-staining of polyGR and AT8



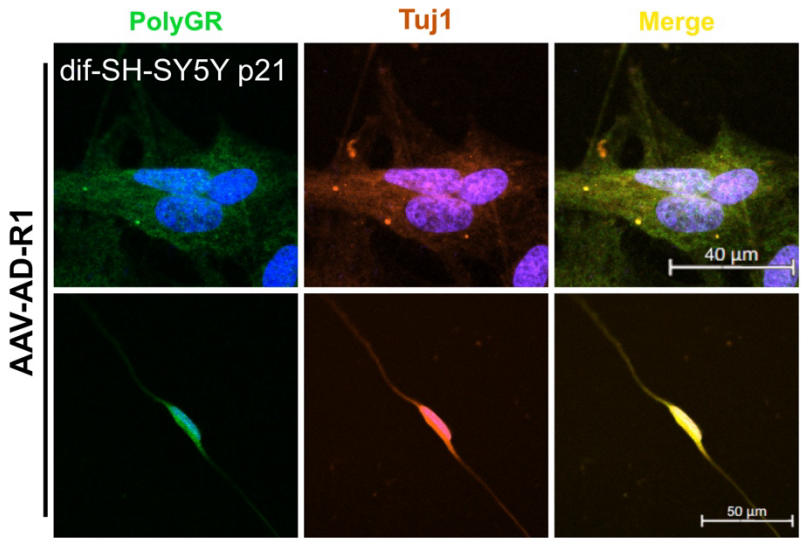
Note. Images taken under 63x magnification. Images are representative of n = 2 cultures per condition. Cells were differentiated for 21 days. Columns 1 and 2 are representative images from the AD-R1 group. Columns 3 and 4 are representative images from the C-Var group. Columns 5 and 6 are representative images from the control group.

PolyGR and Tuj1 staining

A separate set of cells was stained for polyGR and Tuj1 to observe if polyGR can cause changes in morphology. Images of the AD-R1 group were taken. Preliminary qualitative analysis showed that no consistent differences in morphology could be observed compared to the control in Figure 4A. This is shown in Figure 6. Note that results are considered preliminary.

Figure 7

Immunofluorescence co-staining of polyGR and Tuj1



Note. Images taken under 63x magnification. Cells were differentiated for 21 days. Representative images of polyGR and Tuj1 staining in the AD-R1 group. Results are preliminary.

Discussion

Brightfield imaging confirms differentiation. Brightfield images taken during the differentiation procedure show differences in branching (Figure 4A), clustering (Figure 4B), and cell development that suggest differentiation is ongoing.

IF staining confirms effective differentiation. IF staining of MAP2 in RA-differentiated SH-SY5Y cells demonstrated significant branching and synaptic development (Figure 5B). This neuron-type specificity was further shown in staining of Figure 5C, with glutamate expression indicating differentiation into excitatory neurons. Qualitative analysis confirmed that both AAV-transduced (Figure 5A) and non-transduced (Figure 5B) cells yielded branching that indicated maturation. This indicates that our differentiation protocol was overall effective.

PolyGR expression confirms effective AAV transduction. IF staining of polyGR in AAV-C8-AD-R1, AAV-C-Var, and AAV-Cntl yielded expected results, with AD-R1 and C-Var SH-SY5Y cells expressing qualitatively noticeable amounts of polyGR while the control expressed little to no noticeable amounts (Figure 6). This suggests that the $GGGAGA^{exp}$ vector within successfully entered the cells in the AD-R1 and C-Var groups and was expressed.

PolyGR co-localizes with phosphorylated tau (AT8). Based upon qualitative analysis, polyGR was co-localized with AT8 staining in both the AD-R1 and the C-Var groups. In control cells, no co-localization of polyGR and AT8 was observed, nor was expression of polyGR (Figure 6). It should be noted, however, that due to scheduling constraints, not enough images were taken (52 images, n = 2) to perform meaningful statistical analysis. Nonetheless, the visual difference between co-localization of $GGGAGA^{exp}$-transduced groups and the control suggests that polyGR could lead to a putative significant increase of pTau in differentiated SH-SY5Y cells.

SH-SY5Y models could inform future research. This study suggests that SH-SY5Y cells could be informed models suitable for further research on further investigation of AD. Other research topics of interest, such as other repeat expansions mutations related to AD or other neurological disorder research, could also utilize SH-SY5Y models. Because of the cancerous

nature of un-differentiated SH-SY5Y, these cells are easily accessible, viable, and relatively resistant to environmental changes. We show that a 21-day differentiation protocol utilizing RA can successfully lead them to grow into mature neuron-like cells, making them effective *in vitro* models for neuroscience research.

Limitations and future directions. Because we utilized a cellular model, we could only effectively investigate intracellular mechanisms through IF staining. Extracellular accumulation of Aβ, for example, would have been washed off in the process of IF. Therefore, future experimentation has yet to investigate the link between polyGR and Aβ. Western blotting could be utilized in the future to quantify Aβ expression as well as the Aβ42/Aβ40 ratio to examine plaque formation. Due to various time, scheduling, and experimental constraints, not enough images were taken for statistical analysis to be meaningful. Qualitative observations were relied upon, which inherently does not contain statistical power. Furthermore, neuron-specific staining was not performed. Staining for specific neurons, like cholinergic or dopaminergic, or shapes, like bipolar, pyramidal, or unipolar, would have provided evidence towards whether further differentiation specificity could be reached. Staining for specific neuron types could also help establish a relationship with neurotransmitters, polyGR, and AD-NCs. Next, only polyGR was stained for, although C8-AD-R1 can produce poly-glycine-glutamate (polyGE) and poly-arginine-glutamate (polyRE). These could be stained for to investigate their relationship with AD-NCs. Finally, further solidification of findings could encompass different staining methods, such as immunohistochemistry (IHC) staining or RNA Fluorescence *In Situ* Hybridization (RNA-FISH) staining.

Conclusion

We found that our adapted differentiation protocol using RA, FBS, and various other growth factors was successful in yielding mature, differentiated cells, as they expressed mature neuronal markers. Morphological traits resembled mature neurons; this was confirmed through qualitative analysis of brightfield images taken. Through qualitative analysis of IF staining for polyGR and pTau, we observed significantly co-localization of polyGR and AT8 in AD-R1 and C-Var compared to very little expression of polyGR and almost no co-localization in the control, suggesting that polyGR aggregates cause or contribute to pTau accumulations. These results provide further evidence that C8-AD-R1 can cause AD-NCs, insight into new cellular models that could be used to observe AD-NCs. Altogether, the study provides further insight into AD's identity as a repeat expansion mutation and emphasizes the importance of mechanisms like RAN translation in neuropathogenesis.

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